

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
APPLIED BUSINESS ANALYTICS PRACTICAL
Practical – IB

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Class:TYIT	Batch:1
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Q2.Analyza customer data to identify trends and pattern that can be used for business decisions making

Import requires libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Load dataset

```
df = pd.read_csv("Retail_sales.csv")
```

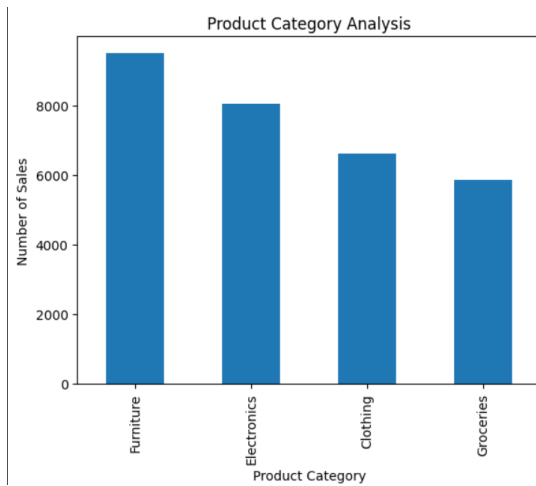
Product category analysis'

```
print(df["Product Category"].value_counts())
```

```
Product Category
Furniture      9503
Electronics    8041
Clothing       6608
Groceries      5848
Name: count, dtype: int64
```

Product category analysis

```
plt.title("Product Category Analysis")
plt.xlabel("Product Category")
plt.ylabel("Number of Sales")
plt.show()
```

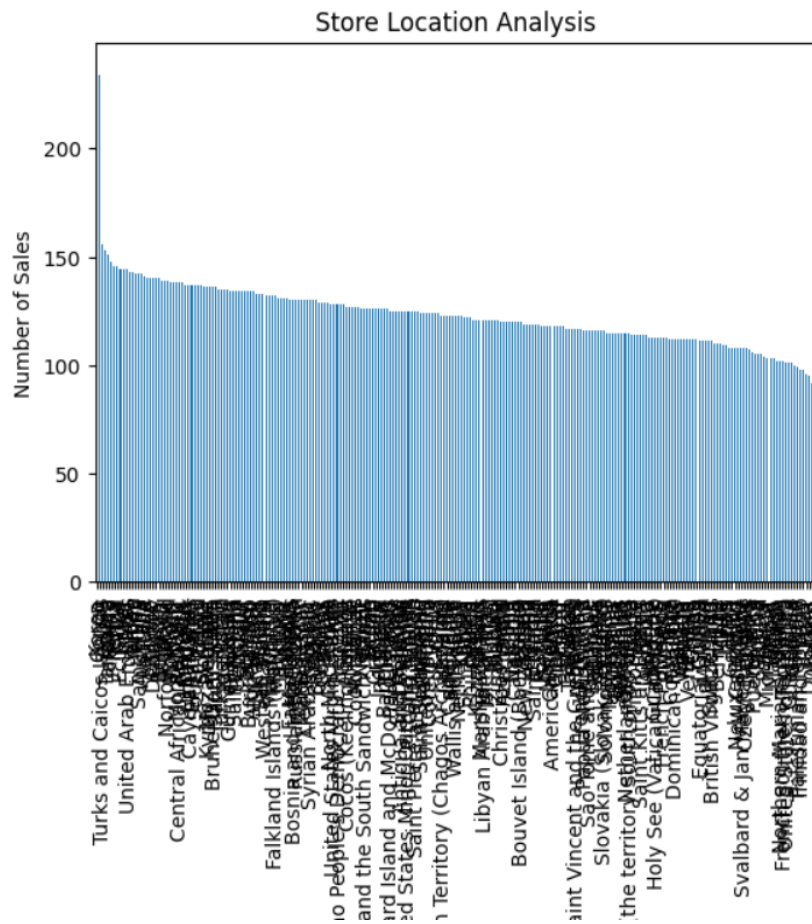


Analysis 2- store location analysis

```
print(df["Store Location"].value_counts())
```

```
Store Location
Korea          237
Congo          234
Turks and Caicos Islands  156
Guam           153
Tanzania       151
...
Trinidad and Tobago  98
Haiti           96
Dominica        95
Ireland         92
Andorra         80
Name: count, Length: 243, dtype: int64
```

Location analysis

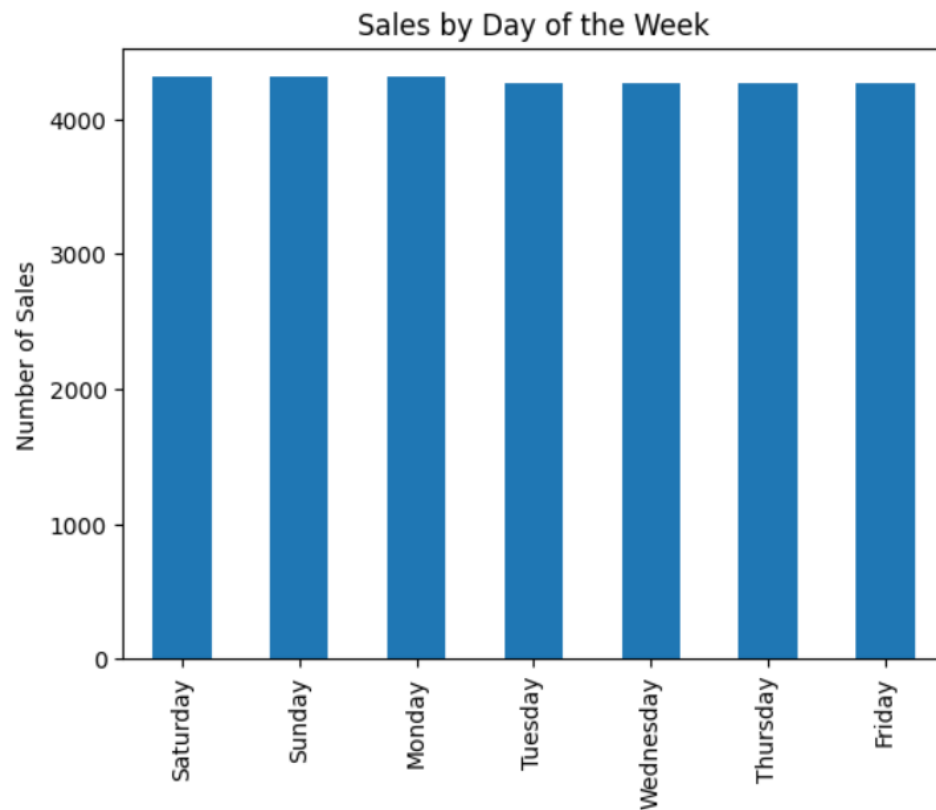


Analysis

```
print(df["Day of the Week"].value_counts())
```

```
Day of the Week
Saturday      4310
Sunday        4309
Monday        4309
Tuesday       4268
Wednesday     4268
Thursday      4268
Friday        4268
Name: count, dtype: int64
```

```
df["Day of the Week"].value_counts().plot(kind="bar")
plt.title("Sales by Day of the Week")
plt.xlabel("Day")
plt.ylabel("Number of Sales")
plt.show()
```

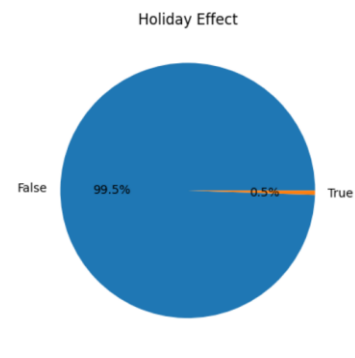


Analysis day 4

```
print(df["Holiday Effect"].value_counts())
```

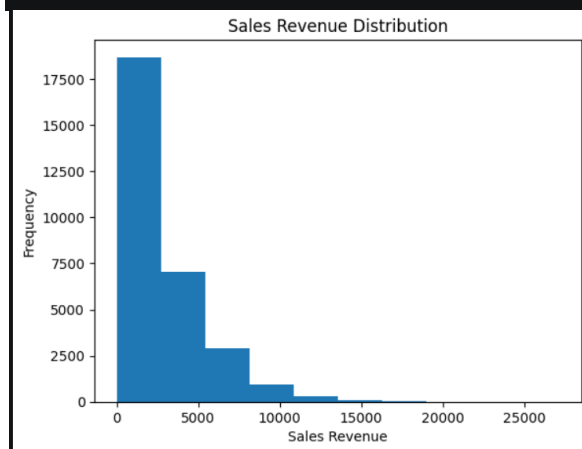
```
Holiday Effect
False    29836
True      164
Name: count, dtype: int64
```

```
df["Holiday Effect"].value_counts().plot(kind="pie", autopct="%1.1f%%")
plt.title("Holiday Effect")
plt.ylabel("")
plt.show()
```



Day 5 sales analysis

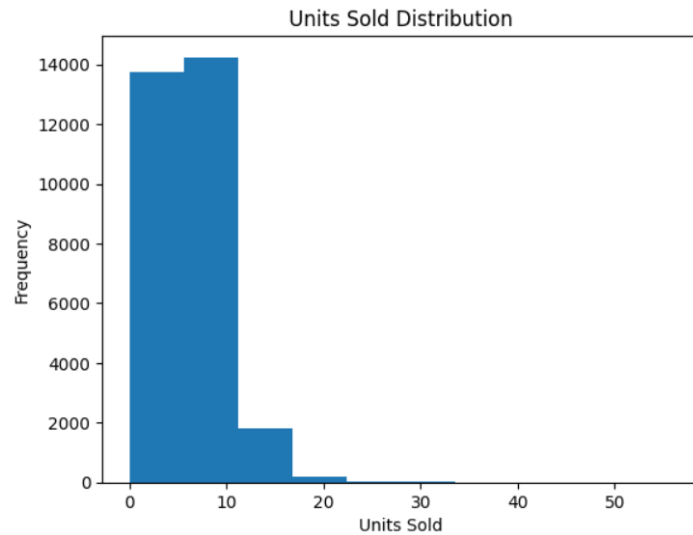
```
plt.hist(df["Sales Revenue (USD)"], bins=10)
plt.title("Sales Revenue Distribution")
plt.xlabel("Sales Revenue")
plt.ylabel("Frequency")
plt.show()
```



Analysis 6 units sold analysis

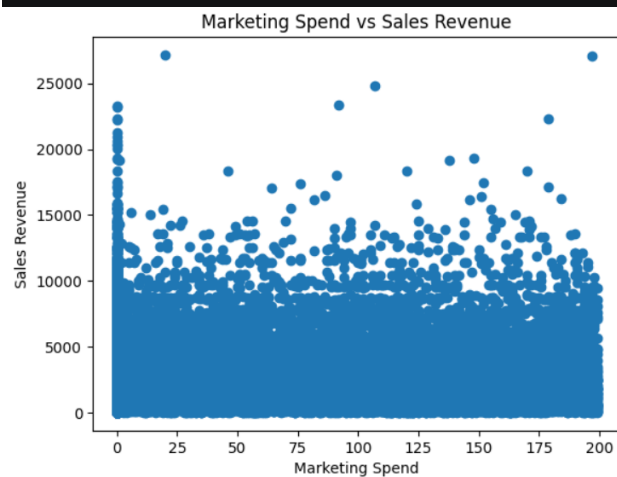
```
plt.hist(df["Units Sold"], bins=10)
```

```
plt.title("Units Sold Distribution")
plt.xlabel("Units Sold")
plt.ylabel("Frequency")
plt.show()
```



Analysis 7 marketing sales

```
plt.scatter(df["Marketing Spend (USD)"], df["Sales Revenue (USD)"])
plt.title("Marketing Spend vs Sales Revenue")
plt.xlabel("Marketing Spend")
plt.ylabel("Sales Revenue")
plt.show()
```



Analysis 8 correlation analysis

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Oranges")  
plt.title("Correlation Heatmap")  
plt.show()
```

